

AI by Hand 🖐️

Agentic AI Problem Set

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in collaboration with

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Dear Readers,

I developed this problem set for my students to study the 20 key terms for agentic AI that became popular in 2025. I want to share this problem set with you.

In the spirit of AI by Hand 🍌, here is the design philosophy behind this problem set:

1. **Problem-Driven.** Every term begins with a problem statement—what motivated this idea in the first place.
2. **Think-First:** Instead of giving away the term immediately, you first get a multiple-choice question so you can *think* about which idea (term) best solves that problem.
3. **Solution:** On the next page, I reveal the correct term and explain how it addresses the problem.
4. **Industry Insights:** As a university professor, I don't have industry experience. I'm fortunate to collaborate with Ofer Mendelevitch (Vectara), who contributes real-world industry perspectives.

Hope you enjoy this problem set! 😊

~ Prof. Tom Yeh



20 Core Agentic AI Concepts Review

1. **LLM Agent** — Uses an LLM's general reasoning ability to follow instructions.
2. **Tool Use** — Lets agents take real actions instead of just explaining steps.
3. **Function Calling** — Ensures tool calls are structured and machine-executable.
4. **Reason-and-Act (ReAct)** — Makes agents reason explicitly before taking actions.
5. **Chain of Thought** — Encourages step-by-step reasoning instead of shallow answers.
6. **Agent Loop** — Adds continuous feedback so agents can observe outcomes and adjust actions.
7. **Reflection** — Allows agents to review mistakes and correct themselves.
8. **Critic** — Provides preference signals to judge which outputs are better.
9. **Plan-and-Execute** — Separates planning from execution for structured task completion.
10. **Episodic Memory** — Stores short-term events and recent actions.
11. **Semantic Memory** — Stores long-term knowledge so expertise carries over.
12. **Context Selection** — Chooses only relevant information for the prompt.
13. **RAG (Retrieval-Augmented Generation)** — Grounds reasoning in external data.
14. **Tree of Thought** — Generates and evaluates multiple reasoning branches in parallel.
15. **Graph of Thought** — Links and reuses insights across those branches to support non-linear reasoning.
16. **Graph RAG** — Reasons over relationships between retrieved documents.
17. **Delegation** — Hands off work to tools or sub-agents to avoid bottlenecks.
18. **Orchestration** — Coordinates tools, results, and execution across steps.
19. **Model Control Protocol (MCP)** — Standardizes secure, auditable access to tools and data.
20. **Safety Guardrails** — Enforces boundaries for safe, governed agent behavior.

Problem 1 of 20

Agents blindly move forward without continuous feedback to adjust their actions.

Solution?

(1) Function Calling	(2) Agent Loop
(3) Reason-and-Act	(4) Tree of Thought

Answer: (2)

Agent Loop

Agents blindly move forward if they don't receive continuous feedback to adjust their actions. The Agent Loop fixes this by making the agent pause after each step, look at what happened, and adjust its next move.

Industry Insight

“Most agents today are simple tool loops — but making them production-grade is the real challenge”

Ofer Mendelevitch

Author, Hands-On RAG for Production, O'Reilly

Head of Developer Relations, Vectara



Problem 2 of 20

Agents cannot connect insights across multiple reasoning branches.

Solution?

(1) Delegation	(2) Chain of Thought
(3) Graph of Thought	(4) Context Routing

Answer: (3)

Graph of Thought

Agents think in a straight line and miss relationships between different ideas. Graph of Thought fixes this by letting the agent explore several branches and link them together, so it can combine insights instead of losing them.

Problem 3 of 20

Agents struggle to keep tools, results, and steps in sync.

Solution?

(1) Delegation	(2) Chain of Thought
(3) RAG	(4) Orchestration

Answer: (4)

Orchestration

Agents can struggle to coordinate tools, intermediate results, and execution steps. Orchestration fixes this by executing tool calls, passing results forward, and iterating until completion.

Problem 4 of 20

Agents access tools and data in inconsistent ways.

Solution?

(1) Reflection	(2) Context Routing
(3) RAG	(4) Model Control Protocol

Answer: (4)

Model Control Protocol

Agents sometimes access tools and data in ad-hoc, non-standard ways, making everything harder to maintain. MCP fixes this by giving them one clear, permissioned protocol for every action, so all calls follow the same safe, consistent, auditable pattern. It replaces a mess of one-off integrations with a system you can actually maintain.

Problem 5 of 20

Agents cannot reason over relationships between retrieved documents.

Solution?

(1) Model Control Protocol	(2) Delegation
(3) Chain of Thought	(4) Graph RAG

Answer: (4)

Graph RAG

Agents treat retrieved documents as isolated chunks and miss how they relate. Graph RAG solves this by building a graph of connections—people, places, events—so the agent can reason across the whole network, not just one document at a time.

Problem 6 of 20

Agents cannot generalize to new instructions.

Solution?

(1) Context Routing	(2) Graph RAG
(3) LLM Agent	(4) Safety Guardrails

Answer: (3)

LLM Agent

Agents fail on new tasks when they rely on fixed rules instead of real reasoning. The LLM Agent setup fixes this by letting the LLM do what it's best at—general reasoning—while giving it structure, memory, and tools so it can apply that reasoning to follow new instructions with flexibility, not brittle rules.

Problem 7 of 20

Agents cross boundaries they didn't know existed.

Solution?

(1) Agent Loop	(2) Graph RAG
(3) Safety Guardrails	(4) Delegation

Answer: (3)

Safety Guardrails

Agents can wander into unsafe or unwanted behavior simply because they don't know the limits. Safety Guardrails fix this by defining clear boundaries and rules, so the agent understands where it can and cannot go.

Industry Insight

“45% of enterprises will struggle to operationalize agentic prototypes due to governance and infrastructure gaps.”

Ofer Mendelevitch

Author, Hands-On RAG for Production, O'Reilly

Head of Developer Relations, Vectara



Problem 8 of 20

Agents produce correct outputs but have no sense of preference.

Solution?

(1) Semantic Memory	(2) LLM Agent
(3) Critic	(4) Agent Loop

Answer: (3)

Critic

Agents may generate multiple possible actions or responses but lack a mechanism to judge which ones are better. A critic provides this missing preference signal by evaluating outputs against goals, constraints, or quality criteria. This feedback allows the agent to compare options, revise decisions, and iteratively improve its behavior.

Problem 9 of 20

Agents improvise instead of forming a plan.

Solution?

(1) Safety Guardrails	(2) Plan-and-Execute
(3) Agent Loop	(4) Tree of Thought

Answer: (2)

Plan-and-Execute

Agents often jump straight into doing things with no structure. Plan-and-Execute fixes this by making the agent outline a clear plan first, then follow it step by step so the work stays organized and predictable.

Problem 10 of 20

Agents cannot use my own data to support their reasoning.

Solution?

(1) Safety Guardrails	(2) RAG
(3) Graph of Thought	(4) Graph RAG

Answer: (2)

RAG

Agents rely only on what's inside the model and ignore the user's actual documents. RAG fixes this by letting the agent retrieve your data and use it directly in its reasoning, so answers are grounded in your real information.

Problem 11 of 20

The model selects the right tool, but doesn't produce a valid tool call.

Solution?

(1) Tool Use	(2) Reflection
(3) Graph RAG	(4) Function Calling

Answer: (4)

Function Calling

Agents may select the right tool, but the LLM may fail to emit the tool call in a machine-executable format. Function Calling fixes this by requiring the model to follow a strict schema, ensuring tool calls include the correct arguments.

Problem 12 of 20

Agents repeat the same wrong answer every time.

Solution?

(1) Plan-and-Execute	(2) Reflection
(3) RAG	(4) Tool Use

Answer: (2)

Reflection

Agents often make the same mistake because they never pause to review their own work. Reflection fixes this by making the agent look back, spot errors, and correct itself before trying again.

Problem 13 of 20

Agents act with no reasoning behind the action.

Solution?

(1) Reason-and-Act (ReAct)	(2) Reward Model
(3) Semantic Memory	(4) Graph of Thought

Answer: (1)

Reason & Act (ReAct)

Agents sometimes take actions suddenly without explaining why. Reason-and-Act fixes this by making the agent think first and then act, so every action has a clear, visible chain of reasoning behind it.

Industry Insight

“The hard part of agents isn’t the reasoning — it’s everything around it: sessions, tools, state, safety, and scale.”

Ofer Mendelevitch

Author, *Hands-On RAG for Production*, O'Reilly

Head of Developer Relations, Vectara



Problem 14 of 20

Agents give shallow replies that skip important steps.

Solution?

(1) Agent Loop	(2) Semantic Memory
(3) Delegation	(4) Chain of Thought

Answer: (4)

Chain of Thought

Agents may give shallow replies because they jump straight to an answer and skip the steps in between. Chain of Thought fixes this by letting the agent think out loud, breaking the problem into small, clear steps. This slows the agent down just enough to show its reasoning, catch mistakes, and produce answers that are deeper, clearer, and more accurate.

Problem 15 of 20

Agents grab too much irrelevant text into the prompt.

Solution?

(1) Episodic Memory	(2) Semantic Memory
(3) Orchestration	(4) Context Selection

Answer: (4)

Context Selection

Agents may pull in too much text because they lack a way to decide what information is relevant for the current step. Context selection addresses this by deliberately choosing which pieces of retrieved text, memory, or prior interaction enter the prompt. Instead of loading entire documents or full conversation histories, only the most relevant information is included, keeping the prompt focused, compact, and easier for the agent to reason over.

Problem 16 of 20

Agents cannot build expertise that carries over.

Solution?

(1) Reason-and-Act	(2) Semantic Memory
(3) RAG	(4) Safety Guardrails

Answer: (2)

Semantic Memory

Agents often start from scratch because they lack persistent memory across interactions. Semantic memory, a form of long-term memory, addresses this by storing facts, concepts, and distilled lessons, so the agent can retrieve and reuse knowledge over time. This allows the agent to accumulate expertise instead of relearning the same information repeatedly.

Problem 17 of 20

Agents tell me what to do but can't do it for me.

Solution?

(1) Safety Guardrails	(2) Semantic Memory
(3) Tool Use	(4) Graph of Thought

Answer: (3)

Tool Use

Agents often explain the steps but never take the action themselves. Tool Use fixes this by letting the agent call the tools directly, so it can perform tasks instead of handing everything back to you.

Problem 18 of 20

Agents forget what happened five minutes ago.

Solution?

(1) Episodic Memory	(2) Tool Use
(3) Context Routing	(4) RAG

Answer: (1)

Episodic Memory

Agents struggle when they can't remember recent actions or past attempts. Episodic Memory solves this by storing short-term events the agent can look back on, so it doesn't repeat mistakes or lose track of what it was doing.

Problem 19 of 20

Agents bottleneck everything by doing all the work alone.

Solution?

(1) Delegation	(2) Tree of Thought
(3) Orchestration	(4) Graph of Thought

Answer: (1)

Delegation

Agents slow down when they try to handle every step themselves. Delegation fixes this by letting the agent hand off tasks to other tools or sub-agents, so the work moves faster and doesn't get stuck on one overloaded agent.

Problem 20 of 20

Agents never explore other branches of possibilities.

Solution?

(1) Function Calling	(2) Tree of Thought
(3) Safety Guardrails	(4) Reward Model

Answer: (2)

Tree of Thought

Agents often stick to one line of thinking and miss better options. Tree of Thought fixes this by having the agent explore multiple branches and compare them, leading to more reliable and creative answers.

Industry Insight

“Piecemeal Agentic AI solutions take months to implement — and twice as long to stabilize. Enterprises need an end-to-end Agent Operating System designed to avoid common AI pitfalls.”

Ofer Mendelevitch

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